Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/27/2006
PI Name:	Goree, John Ph.D.		
Project Title:	Self-Structuring in Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	319-335-1843
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City:	Iowa City	**State:**	IA
Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	01/01/2005	**End Date:**	03/31/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chait, Arnon	**Contact Phone:**	216-433-3558
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2007, per info from PI stating that grant was reinstated as a new grant ; see "Self-Structuring in Dusty Plasmas (NNX07AD22G)" for new information (10/2007) NOTE: grant reinstated 8/2006; end date now 5/18/2009, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05GA43G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This fundamental research is an experimental study of dusty plasma physics. Ground-based laboratory research are carried out in preparation for future microgravity flight experiments using European instruments on the International Space Station ISS. A dusty plasma, also known as a “colloidal plasma,” consists of small particles of solid matter suspended in a partially ionized gas. The particles have a negative electrical charge Q, due to constant collection of electrons and ions from the surrounding plasma. The particles are trapped in a volume by natural electric fields in a plasma, and they repel one another through a screened-Coulomb interaction. The work will be done in two stages: • Ground-based experiments prior to spaceflight experiments will serve two purposes: to develop scientific methods for spaceflight experiments, and to provide an immediate scientific return. These will be done in cooperation with the German partners. Topics of study for the immediate scientific return include an experimental characterization and comparison to theory of: compressional and shear waves in both solids and liquids, transport processes such as diffusion, viscosity and thermal conduction. • Spaceflight experiments will be carried in one of the following two European facilities: ESA’s PK4 instrument (which has been funded and is now being designed) or the International Microgravity Plasma Facility (IMPF), (which is planned but has not yet been funded). NASA will not be asked to provide any flight hardware. Objectives of future microgravity experiments are to observe and characterize: weak forces which are obscured by the force of gravity, and self-structuring and dynamics of the particle suspension as a result of these weak forces Work done during this reporting period is for ground-based experiments. This project represents the PI’s role in a collaboration with German co-I’s: Professor Alexander Piel of the University of Kiel, and Professor Andre Melzer of the University of Greifswald.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is fundamental scientific research.
Task Progress:	During the reporting period 1/1/2006 - 12/31/2006: * We have so far published 7 papers in refereed journals and we expect that one more will be accepted before the end of the reporting period. * We have given 2 invited talks and 7 contributed talks at major conferences and workshops. * As part of the international collaboration with European scientists, the PI: - Hosted a postdoctoral scientist from Piel’s group. Dr. Oliver Arp performed an experiment in the PI’s lab July – October 2006, and is now writing a paper with the PI. - Co-authored a book chapter written jointly with the other German co-I, Professor Andre Melzer of the University of Greifswald. - Visited Melzer’s group at the University of Greifswald, and gave a series of 7 hours of lectures. During this visit Melzer and the PI planned future research collaboration. * We adapted our ground-based hardware to produce the newly-developed "Coulomb ball" 3D suspensions of dust in plasmas. This will allow future experiments that more nearly mimic the 3D suspensions that are attained under microgravity conditions. In a Coulomb ball, the force of gravity is mostly cancelled by an externally-applied thermophoretic force, which makes use of a gradient in gas temperature. * We developed new experimental methods for heating 2D suspensions of dust, using the radiation pressure force applied by moving laser beams. These methods will be useful in future spacelight experiments. * We developed new data analysis methods to more precisely measure particle positions in all of our experiments. As always, we aim for a significant scientific impact. It is notable that our publications for 2006 include three papers in Physical Review Letters. Our previous papers continue to accumulate large numbers of citations in the scientific literature.
Bibliography Type:	Description: (Last Updated: 01/23/2025)
Abstracts for Journals and Proceedings	Goree J. "Diffusion and super-diffusion in strongly-coupled dusty plasmas." Invited talk presented at the American Physical Society 48th Annual Meeting of the Division of Plasma Physics, Philadelphia, Pennsylvania, October 30–November 3 2006. Bulletin of the American Physical Society, to be published Nov. 2006. , Nov-2006
Abstracts for Journals and Proceedings	Goree J. "Wave Experiments in Dusty Plasmas: Linear and Nonlinear." Invited talk presented at the American Physical Society April Meeting, Dallas, Texas, April 22-26, 2006. Bulletin of the American Physical Society, 2006 Apr. , Apr-2006
Abstracts for Journals and Proceedings	Goree J. "Dusty Plasmas that Behave like Liquids or Solids." Invited Talk presented at the European Physical Society Conference on Plasma Physics, Rome Italy, June 2006 Distributed at European Physical Society conference, June 2006. , Jun-2006
Articles in Peer-reviewed Journals	Liu B, Goree J, Vaulina OS. "Test of the Stokes-Einstein relation in a two-dimensional Yukawa liquid." Phys Rev Lett. 2006 Jan 13;96(1):015005. Epub 2006 Jan 4. PMID: 16486470 , Jan-2006

Articles in Peer-reviewed Journals	Nosenko V, Goree J, Skiff F. "Bispectral analysis of nonlinear compressional waves in a two-dimensional dusty plasma crystal." Phys Rev E Stat Nonlin Soft Matter Phys. 2006 Jan;73(1 Pt 2):016401. Epub 2006 Jan 3. PMID: 16486282 , Jan-2006
Articles in Peer-reviewed Journals	Donkó Z, Goree J, Hartmann P, Kutasi K. "Shear viscosity and shear thinning in two-dimensional Yukawa liquids." Phys Rev Lett. 2006 Apr 14;96(14):145003. Epub 2006 Apr 13. PMID: 16712085 , Apr-2006
Articles in Peer-reviewed Journals	Nosenko V, Goree J, Piel A. " Laser method of heating monolayer dusty plasmas." Physics of Plasmas, 2006 Mar;13(3):article no. 032106. , Mar-2006
Articles in Peer-reviewed Journals	Piel A, Nosenko V, Goree J. "Laser-excited shear waves in solid and liquid two-dimensional dusty plasmas." Physics of Plasmas, 2006 Apr;13(4):article no. 042104. , Apr-2006
Articles in Peer-reviewed Journals	Vaulina OS, Vladimirov SV, Repin AYu, Goree J. "Effect of electrostatic plasma oscillations on the kinetic energy of a charged macroparticle." Physics of Plasmas, 2006 Jan;13(1): article no. 012111. , Jan-2006
Articles in Peer-reviewed Journals	Nosenko V, Goree J, Piel A. "Cutoff wave number for shear waves in a two-dimensional Yukawa system (dusty plasma)." Phys Rev Lett. 2006 Sep 15;97(11):115001. Epub 2006 Sep 12. PMID: 17025892 , Sep-2006